Lab 31 — Windows Networking and Repair Commands

> **Course:** CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)
> **Topic 06:** Operating Systems — Core 2, 28% of Core 2
> **Exam objective:** Use Windows networking and repair commands to diagnose connectivity and repair system file corruption (Core 2 objective 1.2).

Goal

You run the diagnostic and repair commands that appear directly in the Core 2 objectives, interpreting the output of each rather than merely executing it — because in support work the output is the diagnosis.

What you'll produce

A command output interpretation record and a completed system file integrity check with results explained.

Tools and equipment

Windows PC, Command Prompt as administrator, PowerShell

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Display the full network configuration and record the IP address, subnet mask, default gateway, DNS servers, DHCP server and MAC address.

```
ipconfig /all
```

2. Release and renew the DHCP lease, recording the address before and after to confirm the

DHCP server responded.

```
ipconfig /release && ipconfig /renew
```

3. Display and then clear the DNS resolver cache, and record when clearing it is the correct fix — after a DNS record changes but the old answer is still cached.

```
ipconfig /displaydns | more && ipconfig /flushdns
```

4. Test the loopback address first to confirm the TCP/IP stack itself is working before testing anything external.

```
ping 127.0.0.1
```

5. Test the default gateway to separate a local network fault from an internet fault.

```
ping -n 4 %GATEWAY%
```

6. Test an external IP address and then an external name, and record that success on the IP but failure on the name isolates the fault to DNS.

```
ping -n 4 8.8.8.8 && ping -n 4 www.tertiarycourses.com.sg
```

7. Trace the route to a destination and identify where latency increases sharply or where the path stops.

```
tracert -h 12 www.tertiarycourses.com.sg
```

8. Query DNS directly and record the resolver used and the addresses returned.

```
nslookup www.tertiarycourses.com.sg
```

9. List active connections with the owning process ID so you can identify what is actually using the network.

```
netstat -ano | findstr ESTABLISHED
```

10. Run the system file checker to scan and repair protected system files, and record the result it reports.

```
sfc /scannow
```

11. Repair the component store that sfc itself depends on, and record why DISM must be run first when sfc reports it cannot fix the files.

```
DISM /Online /Cleanup-Image /RestoreHealth
```

12. Check the disk for file system errors, and record the difference between /f which fixes errors and /r which also locates bad sectors and takes far longer.

```
chkdsk C: /scan
```

Test it — verification

Every command produces output you have interpreted in writing; ipconfig /all output identifies all six required fields; you correctly state that ping to IP succeeding while ping to name fails means DNS; and the sfc result is recorded with its meaning.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the `apt-get install` step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against `labs/tools.md`. All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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